# Precise AI – Privacy-Preserving Retrieval-Augmented Generation (RAG) Workflow for Financial Data Analysis

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***Abstract* :** Precise AI is a powerful financial data analysis platform that uses a combination of RAG and privacy-preserving algorithms to get very accurate, easy-to-understand, and secure insights from huge amounts of financial data. To deal with the growing problems of reading unstructured financial data, the system uses document embeddings, semantic search, and big language models to get, mask, analyze, and compute sensitive numerical information. Embeddings are used to make user documents and queries more precise when searching for data. Entity extraction, on the other hand, protects privacy by hiding sensitive or personally identifiable information during processing. The LLM creates dynamic analytical results by looking at masked data, finding relevant functions or computations, and making conclusions based on the data that is available.The system keeps getting better with the addition and storage of new features. When the calculations are finished, the data is combined to produce an analytical conclusion that can be safely scaled and unmasked as needed. Precise AI speeds up financial decision-making by automating anomalies, cash flows, risk assessments, statement analysis, and compliance alerts.The size and amount of financial data can overwhelm most people; it is usually unstructured and, as a result, is difficult to analyze for a person looking for specific information. Therefore, many users will experience delays and mistakes trying to find specific information using this method. Precise AI organizes all financial data, extracts key points, and simplifies the financial data into a more user-friendly, safe and quick analysis that does not require a user to go through each page.

**Keywords :** Document Embeddings; Large Language Models (LLM); Retrieval-Augmented Generation (RAG); Financial Data Analysis.

## I. Introduction

Traditional financial risk analysis is still laborious and prone to mistakes, even with the growing use of generative AI to enhance business processes. Large language models and retrieval-augmented generation (RAG) are useful methods for extracting particular data from audit reports and enhancing overall risk assessments [1].

Large language models have improved financial sentiment research, but difficulties, like hallucinations and a lack of factual grounding, continue to limit accuracy. Although multiagent systems and retrieval-augmented generation provide partial solutions, their combined potential is seldom realized. To improve Financial Data Analysis performance and decision-making, the study proposes a single RAMAS design [2].

This study compares financial portfolio building to big language models, few-shot learning, and retrieval-augmented generation (RAG) on real-world financial data. The method uses Llama 3.1-8B to investigate several asset classes and compare baseline LLM performance to RAG-augmented portfolio optimization [3].

This work evaluates financial sentiment using large language models (LLMs), complex sentiment-scoring algorithms, organized data collection techniques, and a methodical categorization of LLM approaches. It also does statistical correlation research to determine how news emotion affects Bitcoin price swings and offers LLM-driven results in a specific case study [4].

This system manages massive amounts of financial data using big data collecting, preprocessing, statistical analysis, and intelligent data mining techniques. It improves accuracy, automates financial operations, and addresses issues such as low intelligence, limited functionality, and disorganized company financial management by utilizing modular components (data collecting, cleansing, integration, analysis, and visualization) [5].

Companies are increasingly utilizing advances in technology such as ai and fintech to find quicker, more comprehensive ways to perform financial analysis across the supply chain, whereas traditional methods put the onus on the company to have the resources and expertise to execute the financial analysis through complex multi-source web data. The objective of this project is to develop an llm-enhanced text mining system that will allow for greater efficiency in performing financial analysis, enable non-professionally trained users to engage in the business of supply chain finance, and provide increased access to financial information [6].

Research has shown that an increase in artificial intelligence (AI) and machine learning across blockchain technology can improve the security, speed and quality of digital transactions over traditional methods It also highlights the

need for new technology architectures that will support increased capabilities for detecting fraud, automating the process of creating smart contracts, and ensuring the integrity of data in today's financial systems [7].

The rapid evolution of LLMs combined with Sentiment-based Machine Learning has created new possibilities for how investors calculate their own Financial Forecasting and make decisions about investments. By understanding the professional capabilities of these technologies, investors can increase the accuracy of market predictions, make informed decisions based on more than just automated algorithms, and promote collaboration between the machine learning world and the human world [8].

Rapid developments in digital financial services have resulted in an increased reliance on text messaging (SMS); however, the unstructured nature of SMS messages presents challenges to effectively tracking a person's finances. Therefore, this research is important as it provides an on-device means of securely and efficiently identifying, retrieving and organizing SMS messages that relate to a person's finances while maintaining a level of privacy [9].

The recent research underscores how increasingly important Generative AI has become within the global financial-services industry due to its capacity to process extensive sets of information; understand context and support particular analyzer processes. These advances demonstrate that to ensure continued innovation regarding the financial services sector, both potential and problems should be thoroughly examined [10].

### 1.1 Overview

This project will aim at creating a sustainable financial data analysis system that will support users to interpret enormous loads of unstructured data. The biggest hurdle with the topic choice is that financial information is usually contained within spread out long PDFs, reports, and statements, and thus it is difficult to acquire accurate information within a short period. The struggles by many users include having to find the information they need at the right time, re-calcing the computation and carrying out manual searches. It addresses this by utilizing a semantic search, document embedding technology, and a retrieval based system which directs the user queries on the most relevant parts of the uploaded files. It also includes safe data management to ensure confidential information is kept. It uses retrieval pipelines, libraries of text processing, and vector databases.

They are divided into vector databases, text processing libraries and retrieval pipelines that refine the complex financial text into digestible bites to enable the system to provide answers to the user request in a clear, accurate and comprehensible way

## II. Literature Review

Advancements in the financial world have led to the combination of large language models and retrieval-augmented generation in the field of artificial intelligence (AI) to facilitate greater conceptualization of context and the minimization of hallucinations, as suggested by Darji, et al [1]. It has been argued that traditional ML and rule based algorithms are ineffective at identifying contextual differences and small anomalies within extensive audit files, which limits their ability to detect financial audit risk. Although LLM's have great language capabilities, they will likely experience difficulties such as hallucinations and low contextual memory when being independently used within financial auditing. This research identifies a significant lacuna in existing knowledge surrounding the construction of domain-specific RAG systems tailored to provide a degree of factual accuracy and traceability for producing financial audit-related reports. This study has identified a conceptual gap, but has not included specific development tools or benchmarks to evaluate the architectural integrity of RAG platform thus limiting its analytical ability through definition of criteria.

According to Kelvin Du et al. [2], the authors have extended their approach in order to incorporate Financial Sentiment Analysis (FSA) along with building a Retrieval-Augmented Multi-Agent System (RAMAS), which combines Retrieval-Augmented Generation (RAG) with collaborative Large Language Model (LLM) agents. Their results demonstrate that cooperation of agents based on roles increases interpretability while reducing bias in single models. Through the use of RAG, factual basis is improved through the addition of financial information; additionally, because agents work together, the reasoning provided through systems that use these two forms of coordination provides greater reliability. However, the only focus of the authors' study was upon accurate sentiment classification, ignoring many potential processing complexities, scalability for high-frequency trading applications, and robustness during periods of market volatility. Further, separation between optimization of retrieval and protocols for communication between agents provides superior ability for architectural integration.

Hajaghaie [3] explore the use of RAG-augmented LLMs for optimizing portfolios and allocating assets. The results find that while pre-trained static LLMs do not have enough knowledge about dynamic financial markets, using current fundamentals to drive retrieval-based decision-making results in significantly enhanced ability to make decisions. Their experimental results from Llama 3.1 appear encouraging; however, the authors indicate that additional out-of-sample validation and stress testing in difficult market conditions need to be conducted. Also, there is limited opportunity to compare quantitatively LLM-based portfolio optimization comprehensively with traditional approaches to portfolio optimization (e.g., mean-variance), thus creating doubt about whether LLMs are truly financially superior to traditional approaches.

According to Chenghao Liu and his colleagues [4], the recent improvements in financial AI have enhanced our ability to utilize large language models (LLMs) to analyze

investor sentiment, interpret news signals, predict market trends, as well as other market data relating to financial performance. They specifically investigate how well LLMs can predict the price of Bitcoin by analyzing sentiment-determined elements that behave in the same manner as the stock price. While finding that LLMs can effectively interpret news and investor sentiment, they have identified various disconnected datasets and methodologies within this subject area. In addition, most studies only evaluate the immediate effects of sentiment on financial transactions, without providing insight into either long-term behavior or any causal relationships between sentiment-related embeddings and changes in asset price. As a result, there are no consistent datasets available to perform an objective analysis of markets experiencing high levels of volatility in their price movements. Therefore, replication of studies or comparison of research results may become difficult.

It has been shown by Sukanth Korkanti et al. [5] that there are many shortcomings in traditional big data financial solutions. While working with large scale data technologies has improved the ability to monitor risk and perform analytics across very broad time spans, most of these solutions do not provide a fully automated infrastructure, a standardized pipeline, or an integrated approach to data acquisition, data preparation, data modeling and data visualization. The lack of integration in the architecture leads to many inconsistency with output and limits the ability for these solutions to be scalable. They also do not incorporate techniques to retrieve augmented reasoning and intelligent contextual understanding, which are two critical components for interpreting financial documents today.

Jiaxing Wang and others [6] present an LLM-supported text mining method for supply chain finance analytics that will help inexperienced decision makers by increasing ease of access and usability; however, this does not resolve issues of large volumes of unstructured data management, real time secure access to this type of information, or masking critical financial information. Without measures to protect the privacy of data being retrieved both when stored in a database as well during the transit process from the database to applications, it is not applicable for enterprise level finance systems.

Hassam Ahmed [7] and others examined the convergence among Artificial Intelligence (AI), Large Language Models (LLMs), and blockchain technology; they suggest a multi-layered intelligent architecture designed for faster transaction processing and more efficient fraud detection.Their findings provide improved ledger-level security (in addition to ledger automation); however, there is no information on a solution(s) that can provide semantic retrieval of masked financial documents or the contextual reconstruction of encrypted data. While data integrity has improved, the retrieval of data from unstructured material is still not solution oriented.

According to Mohamed Abdelsamie and his colleagues [8], using domain specific large scale language models will yield better results than general purpose large scale language models when predicting finance related trends. In addition to noting advancements made in predictions; they also identify limitations with respect to adapting contextually to new situations, dealing with rapidly changing environment, and secure processing of significant amounts of unstructured data. The proposed integration of retrieval processes (using RAG based methods) to access required document(s) with safety measures related to anonymity, while conceptual in nature; still lacks empirical evidence that supports its practical use.

Naveen M K et al. [9] investigate automatic classification of transactions utilizing natural language processing and machine learning techniques in order to develop secure monitoring systems. While their approach to structured SMS financial notifications has been proven effective, it is incapable of supporting enterprise datasets comprising multiple formats/documents. The study lacks the examination of strategies for real-time retrieval and also does not examine the potential to use privacy-preserving embeddings.

In their research, Akshar Prabhu Desai and co-authors [10] examine the use of Generative-AI technologies within enterprise finance to improve both scalability and speed of analytic processes, while recognizing that there are difficulties associated with various issues such as data quality, domain adaptation, and privacy protection. The authors do not describe any specific architectural design for the safe extraction and retrieval of unstructured financial data.

Accordingly, an architecture that has high technical demands comprised of RAG (Retrieval-Augmented Generation), embedding-based encryption, dynamic indexing, privacy masking, and cross-domain financial analytics must exist. The key design requirements of this architecture are traceability of retrieval, reduced rates of hallucination, scalability across heterogeneous data formats, and demonstrated advantages over traditional ML (Machine Learning) and autonomous LLM (Language Model) systems.

## III. PROPOSED METHODOLOGY

This envisioned financial analytics platform is built with a privacy-preserving Retrieval-Augmented Generation (RAG) architecture to create a separation of semantic reasoning activities from the secure computation of numeric values. This platform is composed of five separate but interrelated functional layers: (1) Document ingestion; (2) Embedding / indexing; (3) Privacy-protecting retrieval; (4) Secure computing engine; and (5) Natural language reporting.

When you upload your financial documents, semantic boundary detection occurs to divide your documents into small parts that are more manageable to work with. Each

piece is then converted into a high dimensional vector embedding using a transformer based embedding model. These embeddings function in two ways: semantic indexing for similarity searches and hiding the raw financial text data. Unlike traditional encryption, the embedding transformation means that there will not be any raw financial text instantly stored in an easily accessible spot, therefore decreasing the exposure risk.

Approximate nearest neighbor search will allow the segments to be indexed to facilitate fast retrieval of similar segments during query processing. After entering a query, the user entry will be embedded into vector space, and the k most relevant financial segments will be returned via cosine similarity.

Information will be anonymized before passing it through to a language model by a privacy tokenization module, which will take sensitive numeric fields i.e. Revenue Numbers, Account Numbers and Asset Values and create deterministic symbolic (or tokenized) representations of those specific numeric fields. The mapping of each tokenized value to the actual numeric value will be securely stored in an isolated secure computing module that is separate from the LLM environment.

The LLM processes completely masked information to perform reasoning (analytical functions), pattern recognition (thoughts), and analysis of analytic functions without access to numerical raw values. If computational data is needed (e.g., profit margin, variation, or growth rate), the system generates a structured analytic request and submits it to an encrypted computed engine.

Numerical calculations are carried out by a secure host environment on the basis of the original and unmasked data. Data relating to sensitivity cannot be exposed to an LLM, and only aggregate or normalised findings are provided back to LLM (e.g., triggers or rates). Finally, the LLM provides a clear textual explanation of the results to the user, given their financial query results.

This approach employs retrieval grounding to reduce hallucinations, utilizes masking for controlling data exposure, and employs a vector index for scalable applications. Creating a barrier between semantic reasoning and numerical computation establishes a privacy-focused AI financial analytics pipeline that can be deployed in an enterprise environment.

It also explains the process in more of a philosophical way than as a technical one, with the key architectural elements being presented before the appropriate formal definitions of what they are. While the term "embedding-based encryption" is often used, the term "embedding" does not necessarily guarantee that it will provide any level of cryptographic security. The masking and unmasking methods are described in a narrative style, without any definitions in terms of math or computation. Lastly, there is no information on how analytic functions were produced—whether rule-based, model-learned, or through symbolic execution.

## IV. Block Diagram

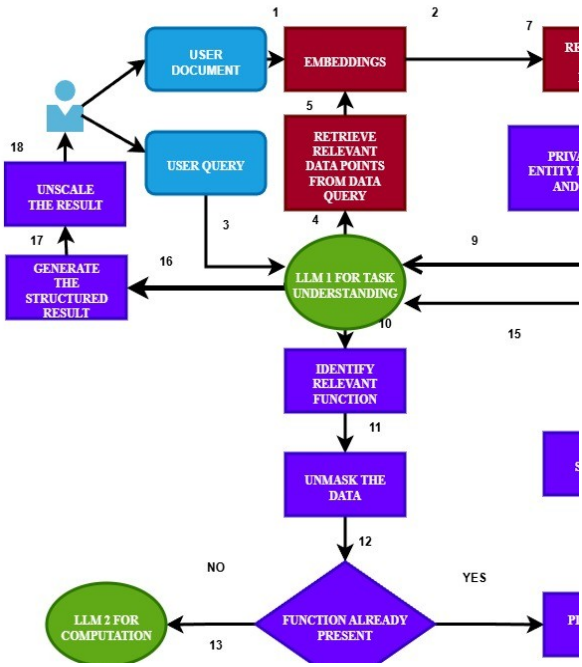


Fig 4.1 : System Architecture Diagram

The block diagram illustrates how a user's financial records and system queries are generally managed.

A document is reviewed and transformed into structured embeddings once the user submits it. When the system needs to locate and match data, these embeddings are stored to help. When the user simultaneously submits a query, the system returns the most pertinent data. Any number must pass a privacy protection process that conceals important information before being used. The system first determines whether the required function is present before beginning any computation.

A number must go through the privacy-protection stage before it will be allowed to use the number within the calculation system. In performing any type of calculation, the first thing the system will do is check if the calculation function is available to be performed. If there is a saved calculation function, the calculation will be performed immediately; otherwise, before proceeding to segment the data points, the system will perform all of the required

calculation on the data. The combined results of the calculations and recovered data will then be combined together for use. At the end of the process, the calculation system will produce a clear and easy-to-understand output for the user.

## V. Result and Discussion

Precise-AI has successfully automated financial document analysis.Referred to collectively, these are the performance metrics of the complete operational flow of the Precise-AI System. Each of the login and signup interfaces utilizes Supabase Authentication and their email verification systems to provide a secure method of providing users access to the features of the Overall System. The Dashboard Interface enables a user to upload documents, view discussion history, as well as response to requests in real-time. The Authentication Console provides evidence to users that they successfully registered for an account as well as successfully integrated to a provider. The Email Verification Screenshot provides evidence of safe activation of the account. All the above represent functional support of the System's overall deployment; secure access control; structured document ingestion and interactive Artificial Intelligence-based document analysis; and that all components of authentication, retrieval and user interfaces have been successfully integrated into a cohesive Financial Document Intelligence Platform.

In addition, Precise-AI accurately predicted many key pieces of information regarding Apple's financial situation from their 2023 Form 10-K, including $29,915,000 operating expenses for 2023. The documentation of the system's actual use shows that the RAG process (Document Embedding, Semantic Search, Query Enrichment, Answer Generation) was done in the correct order, which indicates a consistency with both methodology and technology.
When collecting parts of the original text (PDF) that were semantically similar to the question, it shows that embedding-based indexing is able to weed out answers that do not correspond to the reference source(s) they came from, therefore decreasing the chance of fabricating information.

Historical strategic decision-related inquiries by corporations and/or by companies can be processed through electronic means (i.e., computer system) without requiring the re-uploading of previously stored data. The system provides retrieval persistence and maintains the same context for all data retrieved. The response time associated with using this system is less than or equals (as compared) with reviewing the paper-based records (i.e., documents) through manual methods thus reflecting an improvement in performance. Protection of privacy mechanism (i.e., masking) through processing was established and provided assurance that confidential financial records would remain secure during computation of processing activities conducted under state regulatory authority.

Based on the research findings, RAG-based models of financial analytics have a promising potential to improve auditing and compliance. Query enrichment can enhance user intent pre-search execution thereby increasing the accuracy of results retrieved. In addition, document embeddings improve the contextual mapping of questions to financial segments resulting in more uniform results among various types of questions. However, the performance of broadly defined or open-ended analytical search queries, in general, degraded significantly when document context was not available or when document scanning quality was low. Thus, there are many opportunities for enhancing the contextual modelling and optimisation of OCR preparation.

Ultimately, these results provide evidence that the system offers high applied relevance, operational effectiveness, and privacy/privacy-compliant computational capabilities; establishing it as an acceptable solution for intelligent processing of secure financial documents, along with indicating areas for further enhancement with respect to contextual reasoning abilities.

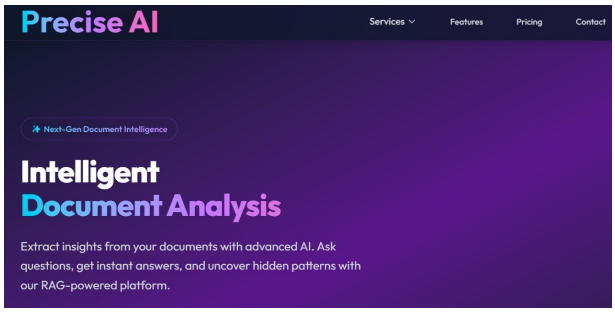


Fig 5.1 : Dashboard

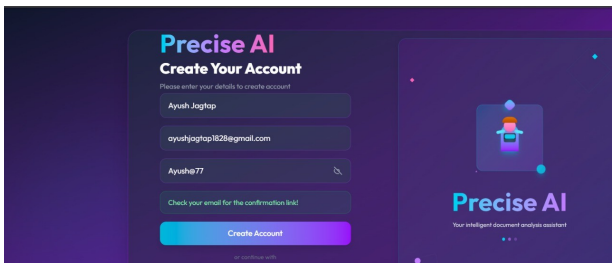


Fig 5.2 : Registration Page

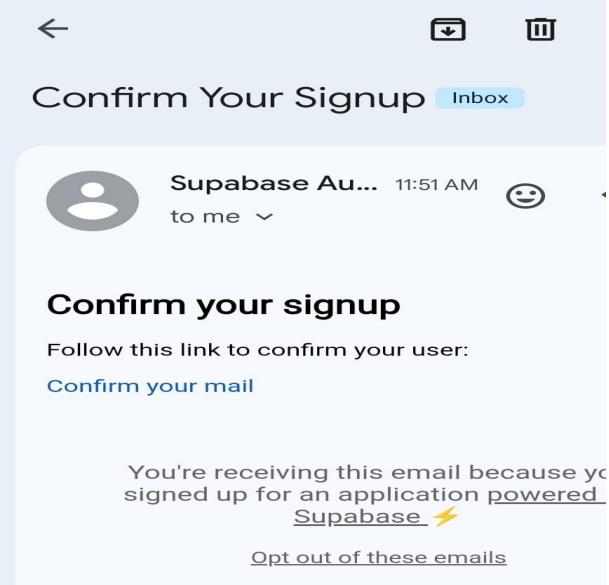


Fig 5.3 : Confirmation Mail Using Supabase

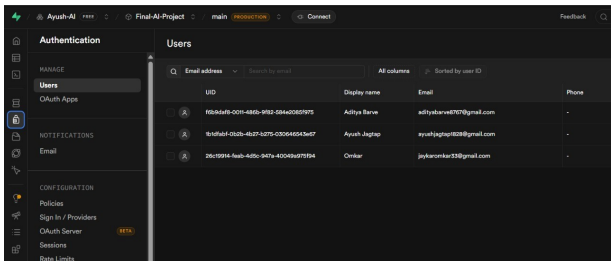


Fig 5.4 : Supabase Authentication

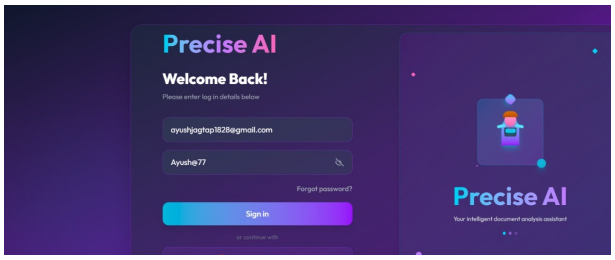


Fig 5.5 : Login Page

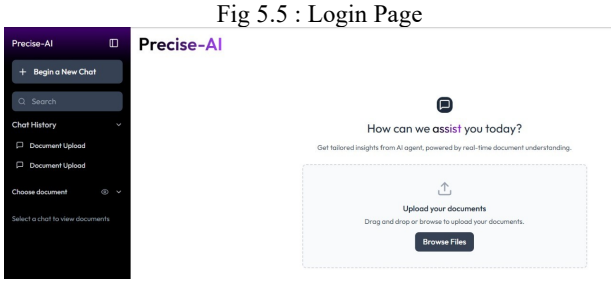


Fig 5.6 : Upload Document

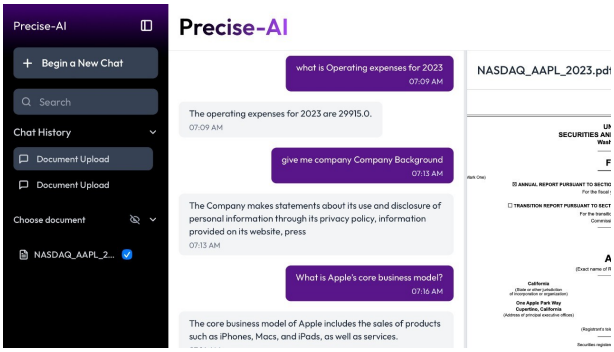


Fig 5.7 : User query for operating expenses and system response

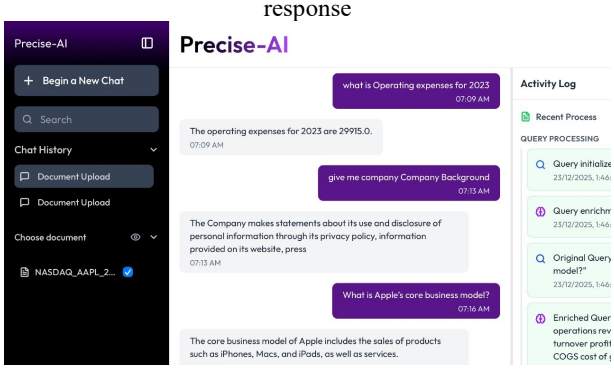


Fig 5.8 :Document retrieval and semantic chunk extraction from Form-10K

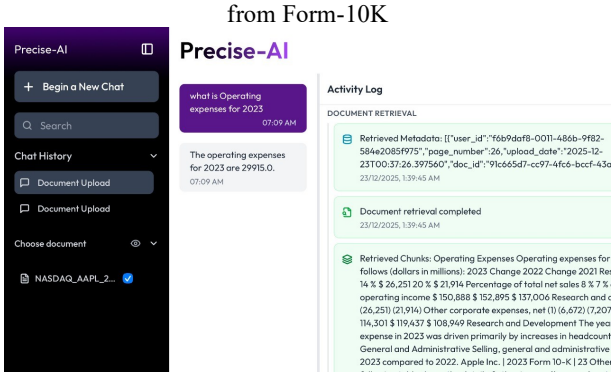


Fig 5.9: Privacy masking and secure data processing activity
log

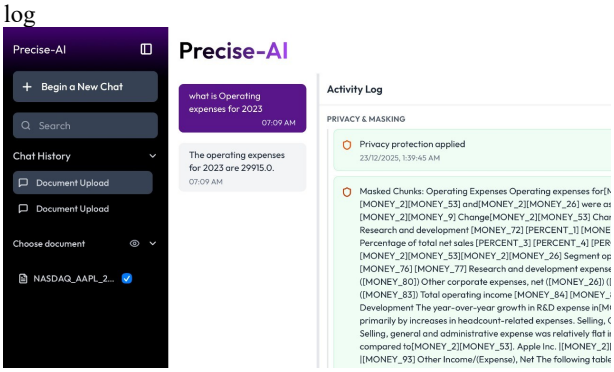


Fig 6.0: Query enrichment and response generation interface

## VI. CONCLUSION

This new data analysis solution combines retrieval of information, generation of structured answers, and

management of documents into one financial analysis system. The results indicate that combining intelligent generation of answers with the method of retrieving information yields higher accuracy, clearer, more reliable results from retrieved financial data. This technology can quickly review unstructured financial records and convert those records into well-organized, usable output, thereby reducing manual labor and human error from the financial reporting process.

Various issues occurred during development: incompatible document formats, both conflicting financial terminology and missing data entries, and both duplicate and/or overlapping querying of each individual document. In addition, another significant challenge was maintaining data privacy and secrecy when working with sensitive numerical data. Achieving this confidentiality, whilst preserving the analytical integrity of the records, required an entire restructuring of the parsing methods, a refining of how queries were interpreted, and the implementation of masking methodologies. Solving these problems led to a considerable improvement in the outcome of the structure, quality, and consistency of records.

There is significant functional depth and breadth demonstrated by the system, as it collects not only critical data but also allocates that data out in a structured and usable format. It is also pointed out in the discussion on the benefits of retrieval based grounding being combined with structured methods of processing to create more reliable and transparent solutions. Ultimately, we want to be able to provide a secure, intelligent and scalable method for managing complex financial records.

Using visual dashboards to graphically display financial data will help the users better interpret and make decisions about the data. Centralized document storage, and the ability to work with many types of documents, will provide easy access and usability for users. Using cloud-based solutions for real-time updates of financial data will ensure users can always work with the most accurate and up-to-date financial information available. The future of financial intelligence services will emphasize developing and enhancing the analytical capabilities and providing automated accuracy while also reinforcing security requirements to create a complete, easy-to-use, and future-proof product.

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